THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET A

Chapters 3 & 4: **The World of Numbers, Exploring Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The decimal expansion of $\frac{7}{20}$ is: 1

(a) terminating **(b)** non-terminating repeating
(c) non-terminating non-repeating **(d)** cannot be determined

2. The decimal expansion of $\frac{13}{25}$ is: 1

(a) non-terminating non-repeating **(b)** terminating
(c) non-terminating repeating **(d)** cannot be determined

3. The number $0.\overline{27}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{3}{11}$ **(b)** $\frac{41}{110}$ **(c)** $\frac{11}{3}$ **(d)** $\frac{4}{11}$

4. The number $\sqrt{12}$ is: 1

(a) rational **(b)** irrational
(c) neither rational nor irrational **(d)** not a real number

5. The number $\frac{22}{7}$ is: 1

(a) not a real number **(b)** rational
(c) neither rational nor irrational **(d)** irrational

6. Expand $(5x + 2)^2$: 1

(a) $25x^2 + 10x + 4$ **(b)** $25x^2 + 4$ **(c)** $25x^2 + 20x - 4$ **(d)** $25x^2 + 20x + 4$

7. Expand $(5x - 1)^2$: 1

(a) $25x^2 - 5x + 1$ **(b)** $25x^2 - 1$ **(c)** $25x^2 + 10x + 1$ **(d)** $25x^2 - 10x + 1$

8. The value of 105^2 using a suitable identity is: 1

(a) 11015 **(b)** 11125 **(c)** 11045 **(d)** 11025

9. The expression $49x^2 + 28xy + 4y^2$ can be factorised as: 1

(a) $(7x - 2y)^2$ **(b)** $(7x + 4y)^2$ **(c)** $(49x + 2y)^2$ **(d)** $(7x + 2y)^2$

10. $(x - 6)(x - 3)$ equals: 1

(a) $x^2 - 9x - 18$ **(b)** $x^2 - 9x + 18$ **(c)** $x^2 + 18x - 9$ **(d)** $x^2 - 3x + 18$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find two rational numbers between $\frac{1}{2}$ and $\frac{3}{4}$. 2

12. A spice trader has a loan of ₹850, makes a profit of ₹1,200 and then a loss of ₹450. Write the sum using integers and find his final standing. 2

13. Factorise $x^2 - 7x + 12$ by splitting the middle term. 2

14. Factorise $s^2 - 11s + 24$ by splitting the middle term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A rational number in its lowest form has denominator $2^3 \times 5$. How many decimal places will its expansion have? Explain. 3

16. Prove that the difference of the squares of two consecutive integers equals their sum. 3

17. Explain, using a debt example, why $10 - (-5) = 15$. 3

OR

Verify the distributive law $\frac{1}{2} \times \left(\frac{1}{4} + \frac{1}{3}\right) = \frac{1}{2} \times \frac{1}{4} + \frac{1}{2} \times \frac{1}{3}$. 3

18. Show that $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$ and use it to find $(105)^2 + (95)^2$. 3

OR

Without multiplying directly, find 104×96 and 98^2. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Simplify $\left(\frac{1}{4} - \frac{5}{12}\right) \div \frac{2}{3}$. 5
(ii) Find the sum and the product of $\frac{3}{5}$ and $\frac{7}{10}$.

OR

Case study: Mādhava's series gives $\pi = 4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots\right)$ and Āryabhaṭa gave $\pi \approx 3.1416$. 5
(i) Is π rational?
(ii) Compute $4\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7}\right)$ and compare with 3.1416.
(iii) Name the mathematician who proved π irrational and the year.

20. (i) Prove $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ using $(a+b)(a+b)^2$. 5
(ii) Deduce $(a-b)^3$ and use it to compute 97^3.

OR

Factorise 5
(i) $x^2 - 5x + 6$,
(ii) $r^2 - r - 42$,
(iii) $6x^2 + 7x + 2$,
(iv) $10x^2 - 11x - 6$.

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND